Photoelectric Effect Lab

1. Definition

The photoelectric effect is the phenomenon in which electrons are emitted from a metal surface when light of sufficiently high frequency falls on it. The emitted electrons are called photoelectrons.

2. Principle

When light falls on a photosensitive metal surface, photons transfer their energy to electrons. If the photon energy is greater than or equal to the work function of the metal, electrons can escape from the surface.

Einstein's photoelectric equation:

$h\nu = \phi + K(max)$

Where:

- h = Planck's constant
- ν = frequency of incident light (Hz)
- ϕ = work function of the metal (J)
- K(max) = maximum kinetic energy of photoelectrons (J)

Maximum kinetic energy:

$K(max) = h\nu - \phi$

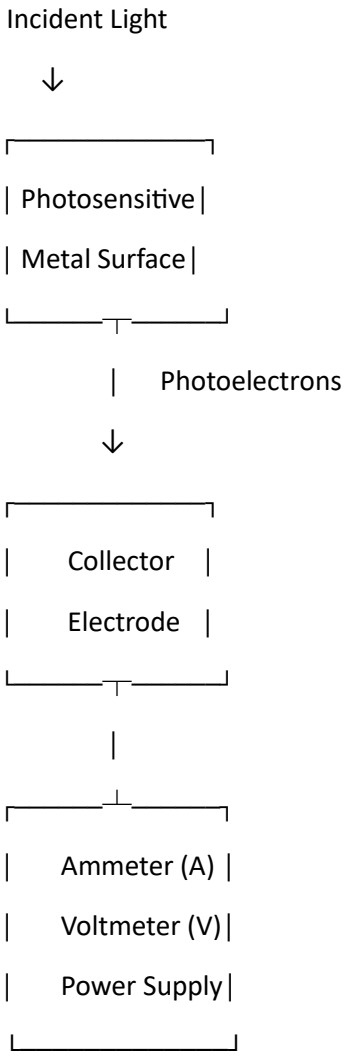
Also,

$K(max) = eV_s$

Therefore,

$eV_s = h\nu - \phi$

3. Experimental Setup



4. Important Terms

- Work function (ϕ): Minimum energy required to remove an electron from a metal surface.
- Threshold frequency (ν_o): Minimum frequency required to produce photoelectric emission.
- Stopping potential (V_s): Minimum reverse voltage required to stop the most energetic photoelectrons.
- Photocurrent: Electric current produced by the emitted photoelectrons.

5. Important Formulae

Threshold frequency:

$\nu_o = \phi/h$

Work function:

$\phi = h\nu_o$

Einstein's equation:

$K(max) = h\nu - \phi$

Stopping potential:

$K(max) = eV_s$

Maximum kinetic energy:

$K(max) = \frac{1}{2}mv(max)^2$

Where:

- $h = 6.626 \times 10^{-34}$ J s
- $e = 1.602 \times 10^{-19}$ C
- m = electron mass

6. Observations

Change in condition	**Observation**
Increase in light intensity (above threshold frequency)	Photocurrent increases
Increase in light frequency	Maximum kinetic energy increases
Frequency below threshold frequency	No photoelectric emission
Increase in positive collector voltage	Photocurrent increases until saturation
Apply sufficient reverse voltage	Photocurrent becomes zero

7. Sample Calculation

Suppose the incident photon energy is 5 eV and the metal's work function is 2 eV.

Using:

$K(max) = h\nu - \phi$

$K(max) = 5 - 2$

$K(max) = 3$ eV

Therefore, the maximum kinetic energy of the emitted electrons is 3 eV.

8. Applications

1. Automatic doors and light sensors.
2. Photocells and light meters.
3. Solar cells (related photon-to-electrical-energy conversion).
4. Smoke detectors.
5. Photoelectric measurement instruments.

9. Result

The photoelectric effect demonstrates that light transfers energy in discrete packets called photons. Photoelectric emission depends on the frequency of incident light, while photocurrent depends on the intensity of light when the frequency is above the threshold frequency.